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Midterm Examination I

Phys331: Electromagnetic Theory I

2026/11/5

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

(1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;

(2) I take *full responsibility* for any ambiguity in my selections in "multiple choice questions";

(3) not selecting any option in "multiple choice questions" will grant 1/8 of the question's points as it shows awareness of lack of knowledge;

(4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;

(5) no partial credit is awarded for the explanations provided in the answer boxes;

(6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded zero points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;

(7) any page which does not contain *both my name and student id* may not be graded;

(8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: _____

Question:	1	2	3	4	5	Total
Points:	16	12	24	24	24	100

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All dimensionful quantities are to be understood in SI units.

(For instance, $\epsilon_0 \simeq 10^{-11}$ and $|\mathbf{B}| = 5$ stand for $\epsilon_0 \simeq 10^{-11}\,\mathrm{A}^2\,\mathrm{kg}^{-1}\,\mathrm{m}^{-3}\,\mathrm{s}^4$ and $|\mathbf{B}| = 5\,\mathrm{A}^{-1}\,\mathrm{kg}^1\,\mathrm{s}^{-2}$.)

Question: 1: Fundamentals of electrostatics ..... *(16 points)*
(Please choose the incorrect statement below)

- ☐ Classical electrodynamic theory is a linear theory.
- ☐ Time independent electric field can only be divergence-free at locations of zero charge density.
- ■ **Electric field being curl-free leads to Kirchhoff's current law.**
- ☐ Principle of superposition is valid in classical electrodynamic theory.
- ☐ Electric field in electrostatics is a conservative vector field.
- ☐ Exact form of Coulomb's law depends on the units system chosen to work with.
- ☐ Electric potential has gauge freedom in its definition.
- ☐ Curl of a vector potential in Helmholtz decomposition for the electric field in electrostatics is zero.

Solution 1.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Electric field being curl free means $\nabla \times \mathbf{E} = 0$. *If we integrate this over an area with boundary curve* C, *we can use Stokes theorem:*

$$\int (\nabla \times \mathbf{E}) \cdot \mathbf{dS} = \oint_C \mathbf{E} \cdot \mathbf{dl} \quad \rightarrow \quad \oint_C \mathbf{E} \cdot \mathbf{dl} = 0 \tag{1}$$

Hence, electric field integrated against a closed loop is zero: this is Kirchhoff's voltage *law.*

Question: 2: Fundamentals of electrostatics ..... *(12 points)*
(Please choose the incorrect statement below)

- ☐ Starting from Gauss law together with $\nabla \times \mathbf{E} = 0$, one can derive Coulomb's law for $\mathbf{E}$.
- ☐ Starting from Coulomb's law for $\mathbf{E}$, one can derive Gauss law.
- ☐ Starting from Coulomb's law for V, one can derive Coulomb's law for $\mathbf{E}$.

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☐ Starting from Coulomb's law for **E**, one can derive Coulomb's law for V.

☐ For a given set of static charge distribution, there is a unique electric field **E** as a function of position.

■ **For a given function $\mathbf{E}(x, y, z)$, there is a unique electric potential V as a function of position.**

☐ For a given function $V(x, y, z)$, there is a unique set of static charge distribution leading to V.

☐ If there were no charges in the universe, $|\mathbf{E}| = 0$ and $V = 0$ would be a consistent solution.

Solution 2.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

We know that $\mathbf{E} = -\nabla V$ *in electrostatics, hence* **E** *does not fix* V *uniquely: there is a gauge freedom at hand!*

Question: 3: Computation of electric potential ..... *(24 points)*

(Modified version of Example 2.8 of 978-1-009-39775-9)

Consider two surfaces S_+ and S_- described by the equations $z = \sqrt{1 - x^2 - y^2}$ and $z = -\sqrt{1 - x^2 - y^2}$ respectively. Assume that charge is distributed uniformly on each surface, with total charge $Q_\pm$ on $S_\pm$, and that no charge flows between the surfaces.

(a) **(12 points)** For $Q_- = 2Q_+ = 4\pi\epsilon_0$, what is the electric potential $V\left(\frac{4}{3}\hat{z}\right)$?

☐ $-3/2$ ☐ -1 ☐ $-1/2$ ☐ $-1/3$ ☐ $1/3$ ☐ $1/2$ ■ 1 ☐ $3/2$

(b) **(12 points)** For $Q_+ = -2Q_- = 4\pi\epsilon_0$, what is the electric potential $V\left(\frac{3}{4}\hat{z}\right)$?

☐ $-3/2$ ☐ -1 ☐ $-1/2$ ☐ $-1/3$ ☐ $1/3$ ☐ $1/2$ ■ 1 ☐ $3/2$

Solution 3.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Squaring the equations $z = \pm\sqrt{1 - x^2 - y^2}$ *leads to* $x^2 + y^2 + z^2 = 1$, *hence the surfaces are parts of a unit sphere. We can then choose*

$$\begin{aligned} x &= \sin\theta\cos\phi \\ y &= \sin\theta\sin\phi \\ z &= \cos\theta \end{aligned} \tag{2}$$

Then we get for $S_\pm$

$$\cos(\theta) = \pm|\cos(\theta)| \tag{3}$$

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hence S_+ and S_- are described by $\theta \in \left(0, \frac{\pi}{2}\right)$ and $\theta \in \left(\frac{\pi}{2}, \pi\right)$ respectively.

Remember Coulomb's law:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV' \tag{4}$$

and by principle of superposition, total potential is sum of individual potentials, hence:

$$\begin{aligned} V(\mathbf{r}) = &\frac{1}{4\pi\epsilon_0} \frac{Q_-}{2\pi} \int_0^{\infty} (r')^2 dr' \int_{-1}^{0} d(\cos\theta') \int_0^{2\pi} d\phi' \frac{\delta(r'-1)}{|\mathbf{r} - \mathbf{r}'|} \\ &+ \frac{1}{4\pi\epsilon_0} \frac{Q_+}{2\pi} \int_0^{\infty} (r')^2 dr' \int_0^{1} d(\cos\theta') \int_0^{2\pi} d\phi' \frac{\delta(r'-1)}{|\mathbf{r} - \mathbf{r}'|} \end{aligned} \tag{5}$$

where we also used

$$\begin{aligned} \rho_+(\mathbf{r}') &= \frac{charge}{area} \delta(r'-1) = \frac{Q_+}{2\pi} \delta(r'-1) \\ \rho_-(\mathbf{r}') &= \frac{charge}{area} \delta(r'-1) = \frac{Q_-}{2\pi} \delta(r'-1) \end{aligned} \tag{6}$$

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For $\mathbf{r} = z\hat{z}$*, this simplifies as*

$$\begin{aligned}
4\pi\epsilon_0 V(z\hat{z}) =& \frac{Q_-}{2\pi}\int_0^{\infty}(r')^2dr'\int_{-1}^{0}d(\cos\theta')\int_0^{2\pi}d\phi'\frac{\delta(r'-1)}{\sqrt{(r'\sin\theta'\cos\phi')^2+(r'\sin\theta'\sin\phi')^2+(z-r'\cos\theta')^2}} \\
&+\frac{Q_+}{2\pi}\int_0^{\infty}(r')^2dr'\int_{0}^{1}d(\cos\theta')\int_0^{2\pi}d\phi'\frac{\delta(r'-1)}{\sqrt{(r'\sin\theta'\cos\phi')^2+(r'\sin\theta'\sin\phi')^2+(z-r'\cos\theta')^2}} \\
=&\frac{Q_-}{2\pi}\int_{-1}^{0}d(\cos\theta')\int_0^{2\pi}d\phi'\frac{1}{\sqrt{(\sin\theta')^2+(z-\cos\theta')^2}} \\
&+\frac{Q_+}{2\pi}\int_{0}^{1}d(\cos\theta')\int_0^{2\pi}d\phi'\frac{1}{\sqrt{(\sin\theta')^2+(z-\cos\theta')^2}} \\
=&Q_-\int_{-1}^{0}du\frac{1}{\sqrt{1-u^2+(z-u)^2}}+Q_+\int_0^{1}du\frac{1}{\sqrt{1-u^2+(z-u)^2}} \\
=&Q_-\int_{-1}^{0}du\frac{1}{\sqrt{(1+z^2)-2zu}}+Q_+\int_0^{1}du\frac{1}{\sqrt{(1+z^2)-2zu}} \\
=&Q_-\int_{(1+z)^2}^{1+z^2}\frac{dv}{-2z}v^{-1/2}+Q_+\int_{1+z^2}^{(1-z)^2}\frac{dv}{-2z}v^{-1/2}=\frac{Q_-}{-z}\sqrt{v}\Big|_{(1+z)^2}^{1+z^2}+\frac{Q_+}{-z}\sqrt{v}\Big|_{1+z^2}^{(1-z)^2} \\
=&\frac{Q_-|1+z|-Q_+|1-z|}{z}+(Q_+-Q_-)\frac{\sqrt{1+z^2}}{z}
\end{aligned} \tag{7}$$

which simplifies as

$$V(z\hat{z}) = \frac{1}{4\pi\epsilon_0}\begin{cases}\dfrac{Q}{z}+\delta Q\dfrac{\sqrt{1+z^2}-z}{z} & z\geq 1\\ Q+\delta Q\dfrac{\sqrt{1+z^2}-1}{z} & z\leq 1\end{cases} \tag{8}$$

for $Q := Q_+ + Q_-$ *and* $\delta Q := Q_+ - Q_-$*. We see that, for* $\delta Q = 0$*, we get the familiar result: system behaves like a point charge outside the sphere and potential is constant (no* $\mathbf{E}$*−field) inside the sphere. For the given values of* z*:*

$$\begin{aligned}
V\left(\frac{4}{3}\hat{z}\right)\Big|_{Q_-=2Q_+=4\pi\epsilon_0} &= \frac{1}{4\pi\epsilon_0}\frac{4Q_+ + 2Q_-}{4}\Big|_{Q_-=2Q_+=4\pi\epsilon_0} = 1 \\
V\left(\frac{3}{4}\hat{z}\right)\Big|_{Q_+=-2Q_-=4\pi\epsilon_0} &= \frac{1}{4\pi\epsilon_0}\frac{4Q_+ + 2Q_-}{3}\Big|_{Q_+=-2Q_-=4\pi\epsilon_0} = 1
\end{aligned} \tag{9}$$

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Question: 4: Electric field and energy ..... *(24 points)*
(Modified version of Example 2.9 of 978-1-009-39775-9)

Noble gases (except Helium) have six electrons in their outermost subshell.

Assume that we discover a new noble-gas tomorrow and denote it $\mathcal{N}$. By removing the outermost subshell of electrons from $\mathcal{N}$, we can create an ion with spherically symmetric charge distribution (for a non-noble-gas atom, removing the outermost subshell would rip electrons from the underneath shells as well, which is why this argument is only valid for noble-gases): such an ion can be modelled as a uniformly charged spherical shell of negligible thickness and radius $r_{\mathcal{N}}$.

(a) **(12 points)** For a single $\mathcal{N}$ ion modelled as above, what is the total energy stored if $r_{\mathcal{N}} = \dfrac{3e^2}{\pi\epsilon_0}$ where e denotes the electron charge?

☐ $\frac{1}{8}$ ☐ $\frac{1}{4}$ ☐ $\frac{1}{3}$ ☐ $\frac{1}{2}$ ☐ 1 ■ $\frac{3}{2}$ ☐ $\frac{4}{3}$ ☐ 8

(b) **(12 points)** Consider two $\mathcal{N}$ ions modelled as above, and take the center of the shells as $\mathbf{r}_1$ and $\mathbf{r}_2$ in some coordinate system. What is the magnitude of the total electric field $\mathbf{E}(\mathbf{r})$ evaluated at $\mathbf{r} = \dfrac{4\mathbf{r}_1 + \mathbf{r}_2}{5}$ for $\dfrac{2}{5}|\mathbf{r}_2 - \mathbf{r}_1| = r_{\mathcal{N}} = \sqrt{\dfrac{3|e|}{\pi\epsilon_0}}$?

■ $\frac{1}{8}$ ☐ $\frac{1}{4}$ ☐ $\frac{1}{3}$ ☐ $\frac{1}{2}$ ☐ 1 ☐ $\frac{3}{2}$ ☐ $\frac{4}{3}$ ☐ 8

Solution 4.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

Electric field for a uniformly charged spherical shell of negligible thickness, of total charge Q, and of radius $r_{\mathcal{N}}$ is

$$\mathbf{E} = \begin{cases} 0 & \text{if} \quad |\mathbf{r}| < r_{\mathcal{N}} \\ \dfrac{Q}{4\pi\epsilon_0}\dfrac{\mathbf{r}}{|\mathbf{r}|^3} & \text{if} \quad |\mathbf{r}| > r_{\mathcal{N}} \end{cases} \tag{10}$$

as it behaves like a point charge outside and has constant potential V inside.

For part (a), we can find the total energy stored in the system as

$$\begin{aligned} \mathcal{E} &= \frac{\epsilon_0}{2}\int_{\mathbb{R}^3} |\mathbf{E}|^2 d^3x = \frac{Q^2}{32\pi^2\epsilon_0}\int_{\mathbb{R}^3\setminus B^3_{r_{\mathcal{N}}}} \frac{1}{|\mathbf{r}|^4} d^3x \\ &= \frac{Q^2}{32\pi^2\epsilon_0}\int_{r_{\mathcal{N}}}^{\infty} r^2 dr \int_{S^2} (d\Omega)\frac{1}{r^4} = \frac{Q^2}{8\pi\epsilon_0}\int_{r_{\mathcal{N}}}^{\infty}\frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon_0 r_{\mathcal{N}}} \end{aligned} \tag{11}$$

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Since we are told that noble gases have six electrons in their outermost subshell, which are removed for the ion in question, we have

$$Q = 6|e| \tag{12}$$

for the electron charge e. Hence, we have

$$\mathcal{E} = \left.\frac{9e^2}{2\pi\epsilon_0 r_{\mathcal{N}}}\right|_{r_{\mathcal{N}}=\frac{3e^2}{\pi\epsilon_0}} = \frac{3}{2} \tag{13}$$

For part (b), we simply use the linearity of the electric field:

$$\mathbf{E}(\mathbf{r}) = \begin{cases} \dfrac{Q}{4\pi\epsilon_0}\dfrac{\mathbf{r}-\mathbf{r}_1}{|\mathbf{r}-\mathbf{r}_1|^3} + \dfrac{Q}{4\pi\epsilon_0}\dfrac{\mathbf{r}-\mathbf{r}_2}{|\mathbf{r}-\mathbf{r}_2|^3} & \text{if} \quad |\mathbf{r}-\mathbf{r}_1| > r_{\mathcal{N}} \ \wedge \ |\mathbf{r}-\mathbf{r}_2| > r_{\mathcal{N}} \\ \dfrac{Q}{4\pi\epsilon_0}\dfrac{\mathbf{r}-\mathbf{r}_1}{|\mathbf{r}-\mathbf{r}_1|^3} & \text{if} \quad |\mathbf{r}-\mathbf{r}_1| > r_{\mathcal{N}} \ \wedge \ |\mathbf{r}-\mathbf{r}_2| < r_{\mathcal{N}} \\ \dfrac{Q}{4\pi\epsilon_0}\dfrac{\mathbf{r}-\mathbf{r}_2}{|\mathbf{r}-\mathbf{r}_2|^3} & \text{if} \quad |\mathbf{r}-\mathbf{r}_1| < r_{\mathcal{N}} \ \wedge \ |\mathbf{r}-\mathbf{r}_2| > r_{\mathcal{N}} \\ 0 & \text{if} \quad |\mathbf{r}-\mathbf{r}_1| < r_{\mathcal{N}} \ \wedge \ |\mathbf{r}-\mathbf{r}_2| < r_{\mathcal{N}} \end{cases} \tag{14}$$

We are given $\mathbf{r} = \dfrac{4\mathbf{r}_1 + \mathbf{r}_2}{5}$ *and* $|\mathbf{r}_2 - \mathbf{r}_1| = \dfrac{5}{2} r_{\mathcal{N}}$*, hence we see that*

$$\begin{aligned} |\mathbf{r}-\mathbf{r}_1| &= \frac{1}{5}|\mathbf{r}_2 - \mathbf{r}_1| = \frac{r_{\mathcal{N}}}{2} < r_{\mathcal{N}} \\ |\mathbf{r}-\mathbf{r}_2| &= \frac{4}{5}|\mathbf{r}_2 - \mathbf{r}_1| = 2r_{\mathcal{N}} > r_{\mathcal{N}} \end{aligned} \tag{15}$$

hence we have

$$\mathbf{E}\left(\frac{4\mathbf{r}_1 + \mathbf{r}_2}{5}\right) = \left.\frac{Q}{4\pi\epsilon_0}\frac{\mathbf{r}-\mathbf{r}_2}{|\mathbf{r}-\mathbf{r}_2|^3}\right|_{\mathbf{r}=\frac{4\mathbf{r}_1+\mathbf{r}_2}{5}} = \frac{25Q}{64\pi\epsilon_0}\frac{\mathbf{r}_1-\mathbf{r}_2}{|\mathbf{r}_1-\mathbf{r}_2|^3} \tag{16}$$

therefore

$$\left|\mathbf{E}\left(\frac{4\mathbf{r}_1 + \mathbf{r}_2}{5}\right)\right| = \frac{25Q}{64\pi\epsilon_0}\frac{1}{|\mathbf{r}_1-\mathbf{r}_2|^2} = \frac{Q}{16\pi\epsilon_0 r_{\mathcal{N}}^2} = \frac{3|e|}{8\pi\epsilon_0 r_{\mathcal{N}}^2} \tag{17}$$

Thus we get the final result

$$\left.\left|\mathbf{E}\left(\frac{4\mathbf{r}_1 + \mathbf{r}_2}{5}\right)\right|\right|_{r_{\mathcal{N}}=\sqrt{\frac{3|e|}{\pi\epsilon_0}}} = \frac{1}{8} \tag{18}$$

Question: 5: Solving Laplace equation ..... *(24 points)*
(Modified version of Example 3.3 of 978-1-009-39775-9)

The Laplacian operator in cylindrical coordinates reads as $\nabla^2 = \dfrac{\partial^2}{\partial\rho^2} + \dfrac{1}{\rho}\dfrac{\partial}{\partial\rho} + \dfrac{1}{\rho^2}\dfrac{\partial^2}{\partial\theta^2} + \dfrac{\partial^2}{\partial z^2}$

Consider two concentric cylinders of radius R_1 and R_2, where both cylinders are of length ℓ. Put a coordinate system such that their symmetry axes coincide with the $z-$axis. The respective electric

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potentials on the surface of the cylinders are given as $V_1(\theta)$ and $V_2(\theta)$ where θ is the azimuthal angle in the cylindrical coordinate system. Work in the limit ℓ goes to infinity.

One can show for arbitrary $V_{1,2}$ functions, the potential between the cylinders takes the form

$$V(\rho,\theta,z)=\int_0^{2\pi}\frac{d\phi}{\alpha_1}\sum_{n\in\mathbb{Z}}e^{in(\theta-\phi)}\begin{pmatrix}\rho^n+\delta_{n0}\log(\rho)\\ \rho^{-n}-\delta_{n0}\log(\rho)\end{pmatrix}^T\begin{pmatrix}R_1^{\alpha_2}+\delta_{n0}\log(R_1) & R_1^{\alpha_3}-\delta_{n0}\log(R_1)\\ R_2^{\alpha_2}+\delta_{n0}\log(R_2) & R_2^{\alpha_3}-\delta_{n0}\log(R_2)\end{pmatrix}^{\alpha_4}\begin{pmatrix}V_1(\phi)\\ V_2(\phi)\end{pmatrix} \tag{19}$$

for some known $\alpha_{1,2,3,4}$.

(a) **(12 points)** What is (α_1,α_2)?

☐ $(-2\pi,-n)$ ☐ $(-2\pi,n)$ ☐ $(-\pi,-n)$ ☐ $(-\pi,n)$ ☐ $(\pi,-n)$ ☐ (π,n) ☐ $(2\pi,-n)$ ■ $(2\pi,n)$

(b) **(12 points)** What is (α_3,α_4)?

☐ $(-n,-2)$ ☐ $(n,-2)$ ■ $(-n,-1)$ ☐ $(n,-1)$ ☐ $(-n,1)$ ☐ $(n,1)$ ☐ $(-n,2)$ ☐ $(n,2)$

Solution 5.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

As we are told to work in the limit ℓ goes to infinity, symmetry dictates that V field can not have z dependence, hence the problem reduces to that of a two dimensional one. We are then supposed to solve

$$\nabla^2 V(\rho,\theta)=0 \tag{20}$$

with given boundary conditions. With the separation of variables

$$V(\rho,\theta)=P(\rho)\Theta(\theta) \tag{21}$$

the identity $\dfrac{\nabla^2 V(\rho,\theta)}{V(\rho,\theta)}=0$ *reads*

$$\rho^2\frac{P''(\rho)}{P(\rho)}+\rho\frac{P'(\rho)}{P(\rho)}=-\frac{\Theta''(\theta)}{\Theta(\theta)} \tag{22}$$

Since there are no common parameters to both sides, this is a constant. Call it n^2, then the equations read

$$\begin{aligned}\Theta''(\theta)+n^2\Theta(\theta)&=0\\ \rho^2P''(\rho)+\rho P'(\rho)-n^2P(\rho)&=0\end{aligned} \tag{23}$$

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These are ordinary differential equations: first one is the complex exponential and the second one is simply an Euler differential equation*, both of which can be trivially solved:*

$$\left.\begin{aligned}\Theta(\theta) =& a_1(n)e^{in\theta} + a_2(n)e^{-in\theta}\\ P(\rho) =& c_+(n)\rho^n + c_-(n)\rho^{-n}\end{aligned}\right\} \quad if\ n \neq 0 \tag{24}$$

and

$$\left.\begin{aligned}\Theta(\theta) =& b_1 + b_2\theta\\ P(\rho) =& b_3 + b_4\log(\rho)\end{aligned}\right\} \quad if\ n = 0 \tag{25}$$

for undetermined coefficients a, b, and c. As θ is a compactified *coordinate, i.e. θ → θ + 2π should not affect the solution, we see that n has to be an integer and $b_2 = 0$. By allowing both positive and negative integers we can combine $e^{\pm in\theta}$ solutions to a single one, hence the most general solution can be written as*

$$V(\rho,\theta) = \sum_{n\in\mathbb{Z}} \Big(c_+(n)\Big(\rho^n + \delta_{n0}\log(\rho)\Big) + c_-(n)\Big(\rho^{-n} - \delta_{n0}\log(\rho)\Big)\Big) e^{in\theta} \tag{26}$$

Let us apply the boundary conditions

$$\begin{aligned} V_1(\theta) =& V(R_1,\theta) = \sum_{n\in\mathbb{Z}} \Big(c_+(n)\Big(R_1^n + \delta_{n0}\log(R_1)\Big) + c_-(n)\Big(R_1^{-n} - \delta_{n0}\log(R_1)\Big)\Big) e^{in\theta}\\ V_2(\theta) =& V(R_2,\theta) = \sum_{n\in\mathbb{Z}} \Big(c_+(n)\Big(R_2^n + \delta_{n0}\log(R_2)\Big) + c_-(n)\Big(R_2^{-n} - \delta_{n0}\log(R_2)\Big)\Big) e^{in\theta}\end{aligned} \tag{27}$$

We can utilize the orthonormality of the complex exponentials, i.e.

$$\int_0^{2\pi} e^{i\theta(n-m)}\frac{d\theta}{2\pi} = \delta_{nm} \tag{28}$$

hence

$$\begin{aligned} c_+(n)\Big(R_1^n + \delta_{n0}\log(R_1)\Big) + c_-(n)\Big(R_1^{-n} - \delta_{n0}\log(R_1)\Big) =& \int_0^{2\pi} e^{-in\phi}V_1(\phi)\frac{d\phi}{2\pi}\\ c_+(n)\Big(R_2^n + \delta_{n0}\log(R_2)\Big) + c_-(n)\Big(R_2^{-n} - \delta_{n0}\log(R_2)\Big) =& \int_0^{2\pi} e^{-in\phi}V_2(\phi)\frac{d\phi}{2\pi}\end{aligned} \tag{29}$$

which can be rewritten as

$$\begin{pmatrix} R_1^n + \delta_{n0}\log(R_1) & R_1^{-n} - \delta_{n0}\log(R_1)\\ R_2^n + \delta_{n0}\log(R_2) & R_2^{-n} - \delta_{n0}\log(R_2)\end{pmatrix}\begin{pmatrix} c_+(n)\\ c_-(n)\end{pmatrix} = \int_0^{2\pi} e^{-in\phi}\begin{pmatrix} V_1(\phi)\\ V_2(\phi)\end{pmatrix}\frac{d\phi}{2\pi} \tag{30}$$

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hence

$$\begin{pmatrix} c_+(n) \\ c_-(n) \end{pmatrix} = \int_0^{2\pi} e^{-in\phi} \begin{pmatrix} R_1^n + \delta_{n0}\log(R_1) & R_1^{-n} - \delta_{n0}\log(R_1) \\ R_2^n + \delta_{n0}\log(R_2) & R_2^{-n} - \delta_{n0}\log(R_2) \end{pmatrix}^{-1} \begin{pmatrix} V_1(\phi) \\ V_2(\phi) \end{pmatrix} \frac{d\phi}{2\pi} \tag{31}$$

As (26) can be rewritten as

$$V(\rho,\theta) = \sum_{n\in\mathbb{Z}} e^{in\theta} \begin{pmatrix} \rho^n + \delta_{n0}\log(\rho) \\ \rho^{-n} - \delta_{n0}\log(\rho) \end{pmatrix}^T \begin{pmatrix} c_+(n) \\ c_-(n) \end{pmatrix} \tag{32}$$

we obtain the final result as

$$V(\rho,\theta) = \int_0^{2\pi} \frac{d\phi}{2\pi} \sum_{n\in\mathbb{Z}} e^{in(\theta-\phi)} \begin{pmatrix} \rho^n + \delta_{n0}\log(\rho) \\ \rho^{-n} - \delta_{n0}\log(\rho) \end{pmatrix}^T \begin{pmatrix} R_1^n + \delta_{n0}\log(R_1) & R_1^{-n} - \delta_{n0}\log(R_1) \\ R_2^n + \delta_{n0}\log(R_2) & R_2^{-n} - \delta_{n0}\log(R_2) \end{pmatrix}^{-1} \begin{pmatrix} V_1(\phi) \\ V_2(\phi) \end{pmatrix} \tag{33}$$

Hence, we have a unique solution for any boundary condition pair $V_{1,2}$.

« « « Congratulations, you have made it to the end! » » »